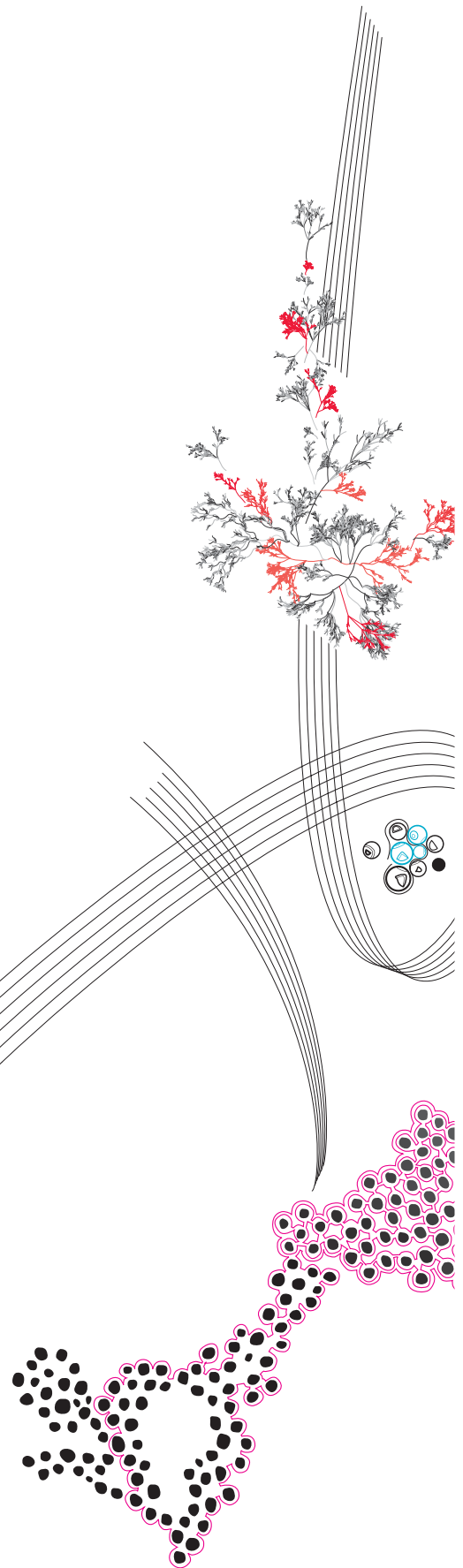




BSc Thesis Creative Technology

Bedside tool for spontaneous bacterial peritonitis diagnosis using impedance spectroscopy

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Abstract

Spontaneous Bacterial Peritonitis (SBP) is a common and life-threatening infection of ascitic fluid in patients with liver cirrhosis. The diagnosis relies on counting neutrophils in ascitic fluid, with a number of ≥ 250 *cells/mm*³ indicating likelihood of infection. Current diagnostic methods are time-consuming and often require manual counting by a trained professional, delaying diagnosis. This project explores the development of a portable, bedside diagnostic tool based on electrical impedance spectroscopy (EIS) for rapid neutrophil counting.

A custom, low-cost impedance-measurement circuit was designed and built around an Arduino, and tested using equivalent circuit models of neutrophils, lymphocytes, and macrophages in PBS solution. Measurements showed a 4–6% difference in impedance between neutrophils and the other cell types, in line with other similar studies. The difference, however, was lower than calculated and measured with a dedicated impedance spectroscope.

Results provide proof of concept of a portable Arduino based system for distinguishing and counting neutrophils in ascitic fluids, however, further work is needed before it can be used in hospitals.

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Chapter 1

Introduction

Patients with liver cirrhosis often develop complications, such as portal hypertension, splanchnic vasodilation and activation of the renin–angiotensin cascade [1]. Dever and Sheikh [1] mention that this leads to sodium and water retention and fluid overflow into the peritoneal cavity. These fluids, called ascitic, provide a favourable environment for growth of bacteria. One of the ascitic fluid infection types is Spontaneous Bacterial Peritonitis (SBP) [1]. Symptoms of SBP can include a presence of fever or hypothermia, chills, abdominal pain and tenderness on palpation. Yet, a third of patients can be completely asymptomatic [2]. Because of that, guidelines recommend that patients with cirrhosis undergo testing for SBP immediately after being admitted to the hospital with any complains. [2]. However, the current diagnosis is time-consuming which can lead to delays in treatment and subsequently, to the death of the patient.

In order to improve the SBP diagnosis process, hospitals, in collaboration with universities, aim to develop a rapid bedside system to diagnose SBP. This project is a part of such collaboration. The aim of the project is to create a bedside lab-on-chip diagnostic system for counting immune cells in a sample of ascites, using Electrical Impedance Spectroscopy (EIS). EIS uses electrodes with excitation voltage to measure impedance as cells flow through a microfluidic channel [3]. These measurements can be used to retrieve various informations about cell properties, such as its size, membrane morphology or information about internal organelles [4]. This way, the cells in the ascitic fluid can be distinguished and counted. The results can be used for SBP diagnosis and do not require much time.

The system will consist of several parts:

- The chip with electrodes through which the fluid sample will flow
- A device that connects to the electrodes and that can take measurements at 2 different frequencies (high and low)
- A screen to display the results

This project is focused on the measurement device.

Chapter 2

Background Research

2.1 Spontaneous Bacterial Peritonitis (SBP)

SBP is an ascitic fluid bacterial infection caused by liver cirrhosis complications like portal hypertension and splanchnic vasodilation. These complications lead to sodium and water retention and fluid overflow into the peritoneal cavity. This is called ascitic fluid, and it is a very favourable environment for bacterial growth [1]. Often, due to the lack of symptoms, the guidelines recommend to test patients with cirrhosis for SBP after they are admitted to the hospital [2].

In order to diagnose SBP, neutrophil count in the ascitic fluid is used. The diagnosis should be considered when the number of neutrophils is $\geq 250/mm^3$ [1],[5]. Dever and Sheikh [1] state that for a proper cell count, 1mL of ascitic fluids needs to be used.

2.2 Flow cytometry

Adan et al. [6] define flow cytometry as a technology for measuring physical characteristics of individual cells as they flow in suspension through a measuring device. In the past, manual neutrophils' counting was the standard for SBP diagnosis [7]. Now, automated hematology analysers based on flow cytometry are becoming more and more popular by being more efficient under the right circumstances [8],[9]. However, in many cases it is still unreliable with ascitic fluids due to the misclassification of non-neutrophils as neutrophils [10]. In these cases, manual count is necessary, which is time consuming and requires the presence of a professional [11].

Electrical Impedance Spectroscopy (EIS), often implemented in microfluidic impedance cytometry systems, provides an alternative label-free approach for single-cell analysis [12]. Instead of working with optical or fluorescence signals, it uses dielectric properties of the cells to detect and distinguish them. This makes it possible to use it for for small, portable equipment [12]. EIS does have some challenges however. They include low sensitivity and a high presence of noise [13]. Gawad et al. [13] contribute these to double-layer effects at the interface of electrodes and electrolyte, channel dimensions (smaller dimensions lead to higher Signal-to-Noise Ratio (SNR), but this can lead to clogging of the cells) and variations in the cell's trajectory. They also mention how the passage of a 5- μ m-diameter cell produces

a signal which corresponds to only a 1% change in the channel impedance, while Honrado et al. [4] mention the challenges with high noise in EIS and how it leads to inaccurate results. There are methods, discussed in a later section, to circumvent these shortcomings, and the portability of EIS devices still make it a great option for developing a bedside SBP diagnostic system.

To better understand EIS, it is important to describe what is being measured and how. Impedance (Z_f) is a frequency-dependent resistance and it is the main output signal in EIS, together with the relative phase. Cells flow individually through a microfluidic channel where pairs of electrodes are attached. Multiple ways of positioning the electrodes exist, the main being coplanar (2D) and facing (3D) [14]. The electrodes have alternating current (AC) flowing through them, with different frequencies connected to each pair of electrodes. This allows measurements at multiple frequencies, which in turn provides information about various cell characteristics. Opacity, defined as ratio of the impedance amplitude at a high frequency to a low frequency ($|Z_{high}/Z_{low}|$), shows the difference in cell resistivity at high and low frequencies. It is used to normalise the data for cell size and distance from the electrodes [3]. For cells whose opacity spectra overlap at a certain frequency range, phase information can be used for differentiation.

2.3 Dielectric properties of biological materials

Since EIS measures the impedance between electrodes, it is important to understand the dielectric properties of biological materials and how they relate to properties of cells. Foster & Schwan [15] define the two basic dielectric properties, permittivity ϵ and conductivity σ as "the charge and current densities induced in response to an applied electric field of unit amplitude" respectively. The authors note that the permittivity and conductivity are frequency dependent and they are related by

$$\sigma^* = j\omega\epsilon^*\epsilon_0 \quad (2.1)$$

where σ^* and ϵ^* are complex conductivity and permittivity respectively, ϵ_0 is permittivity in vacuum and ω is angular frequency. Plotting permittivity over frequency shows three drops of permittivity at different frequencies (Figure 2.1). The regions between the drops are called dispersion regions and Foster & Schwan call them α , β and γ regions. They appear because there are three different polarisation mechanisms, each being active at a characteristic frequency range. Therefore, measurements at different frequencies give information about different cell properties.

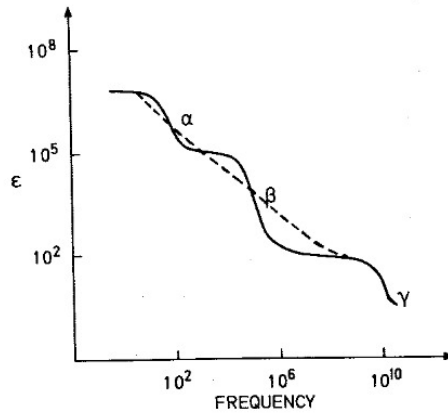


Figure 2.1: Plot of permittivity (ϵ) over frequency.[15]

The α -dispersion lies in the kHz frequency spectrum and is attributed to polarisation of the double layer around the cell [16]. The authors note, however, that it is difficult to measure due to the electrical double layer effects at the interface of electrode and electrolyte, which is dominant at low frequencies. The second, β , dispersion is linked to the charging of the capacitive cell membrane. It occurs in the MHz range and it is widely used to determine the dielectric properties of cells [16]. According to Honrado et al. [4], characteristic sub-dispersions exist for each subcellular region. In the 0.1-1 MHz range, the cell membrane acts as an insulator and the signal amplitude is proportional to the size of the cell. This is also stated by De Ninno et al. [17] and Errico et al. [18]. The membrane starts to polarise at around 1MHz and stays polarised at frequencies in the 1-10 MHz range. At these frequencies it is possible to measure the membrane capacitance which provides information about membrane morphology. Membrane morphology refers to the structural characteristics of the cell membrane, such as its roughness, presence of folds and wrinkles, or protrusions. All of these influence the membrane permittivity, which allows to distinguish two different cells with the same diameter. In the range of 10-30 MHz, the membrane is minimally polarised, so the electric field can go through the cell. The response signal provides information about cytoplasmic conductivity, permittivity and properties of internal organelles [4]. These regions are not strictly defined, so there is an overlap between them. The γ -dispersion lies in the GHz range and is caused by the dipolar relaxation of water.

Since the aim of the project is to count specific cells, the measurements should be taken in the β -dispersion range where cells can be differentiated on multiple properties. For cell detection, measurements at 0.1-0.7 MHz should give the best results, since the membrane does not start to polarise yet. For distinguishing between cells, 1-30 MHz is the ideal range because measurements at these frequencies show the difference in membrane morphology and cytoplasmic properties. Because measurements at too high or too low frequencies negatively impact SNR [4], 0.1-3 MHz is the preferred range to obtain consistent results. Specifically, 300 kHz and 1 MHz were chosen as two measurement frequencies. 300 kHz should be good for cell detection, and 1 MHz allows to distinguish cells based on the membrane morphology.

To study the frequency-dependent behaviour of the cells, the Equivalent Circuit Model (ECM) is used. According to Spencer and Morgan [19] and Sun et al. [20],

the suspending medium is represented with a resistor and capacitor in parallel. The membrane of the cell has a very high resistance, so it can be ignored. Subsequently, the membrane is modelled as a capacitor. The cell cytoplasm is represented by a resistor and a capacitor in parallel, however Sun et al. [20] states that the capacitance can be ignored since in the β -dispersion range its influence is negligible compared to the cytoplasmic conductivity. The resistor in series with the membrane capacitances, as can be seen in Figure 2.2a. Sun et al. [20] additionally introduces the electrode–electrolyte interface impedance, which is modelled as a capacitor (Figure 2.2b). By relating the impedance measurements to the ECM, it is possible to find the values of membrane and cytoplasm impedances. This can be used to distinguish and count the neutrophils.

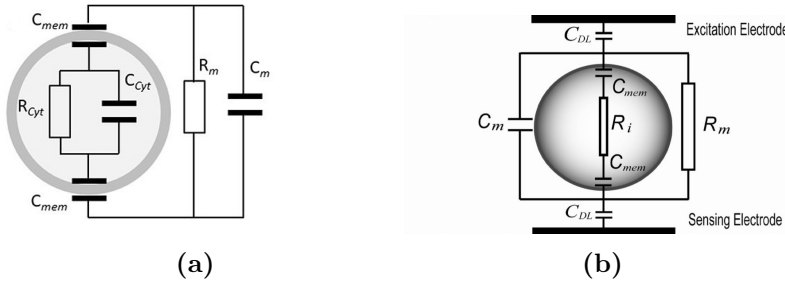


Figure 2.2: Equivalent circuit models for a cell in suspension by [19] (a) and [20] (b)

2.4 Cells in ascitic fluids

For SBP diagnosis, a sample of ascitic fluids is taken from the patients body and the number of neutrophils is counted [1]. However, neutrophils are not the only cells in the ascitic fluids. Besides them, macrophages and lymphocytes are the two most prevalent cell types. Specifically, macrophages make up $28.1 \pm 23.7\%$ of the total cell count and leukocytes $68.4 \pm 25.1\%$ of which $79.8 \pm 20.1\%$ are neutrophils and $16.8 \pm 19.0\%$ are lymphocytes [21]. These are the three most common cell types in ascitic fluids of patients with suspected SBP. Because of that, neutrophils need to be distinguished from macrophages and lymphocytes during the measurement process for accurate diagnosis. For the ECM, $1 \times$ PBS was used as the solution.

Table 2.1 gives the overview of the relevant properties of neutrophils, two types of lymphocytes, macrophages and the solution. The cytoplasmic resistance is calculated using Eq. 2.2 and the membrane capacitance is $C_m = C_{sm} * A_c$ where, $A_c = \pi D_c^2$.

$$R_{cy} = \frac{1}{\sigma_{cy} * \pi * (D_c/2)} \quad (2.2)$$

To calculate resistance of the solution, we take a facing electrodes configuration for simplicity. It can then be assumed that the electric field is uniform. Eq. 2.3 was used to calculate resistance of the solution without the cell in between the electrodes. L is the distance between the electrodes, A is the surface are of the electrodes and

σ_s is the conductivity of the solution.

$$R_s = \frac{L}{\sigma_s A} \quad (2.3)$$

When a cell is in between the electrodes, it displaces a part of the solution, so its resistance changes. The solution resistance with a cell can be calculated using Eq. 2.4 [16]. $K_{cell} = \frac{A}{L}$ is the cell constant and Φ is the volume fraction between the cell and the sensing volume. The equation only works for $D_c \ll L$.

$$R'_s = \frac{1}{\sigma_s(1 - \frac{3}{2}\Phi)K_{cell}} \quad (2.4)$$

With the parameters in Table 2.1 and ECM shown in Figure 2.3, the total output impedance is calculated and 300kHz and 1MHz using Eq. 2.4. Z_{DL} is the double layer impedance, Z_{mem} - membrane impedance, Z_{cy} - cytoplasmic impedance and Z_s is the solution impedance. For macrophages, Eq. 2.6 was used since D_c is not small compared to L . The results are shown in Table 2.2.

$$Z_{tot} = 2Z_{DL} + \frac{(2Z_{mem} + Z_{cy})Z_s}{Z_s + 2Z_{mem} + Z_{cy}} \quad (2.5)$$

$$Z_{tot} = 2Z_{DL} + 2Z_{mem} + Z_{cy} \quad (2.6)$$

TABLE 2.1: Parameters used for simulation circuit

Parameter		Value	Reference
Neutrophils	Diameter, D_c (μm)	10.25	[22]
	Specific membrane capacitance, C_{sm} ($\mu F/cm^2$)	2.17	[22]
	Cytoplasmic conductivity σ_{cy} (S/m)	0.37	[22]
	Cytoplasmic resistance, R_{cy} ($k\Omega$)	168	*
	Membrane capacitance, C_{mem} (pF)	7.16	*
T-lymphocytes	Diameter, D_c (μm)	6.58	[23]
	Specific membrane capacitance, C_{sm} ($\mu F/cm^2$)	1.05	[23]
	Cytoplasmic conductivity σ_{cy} (S/m)	0.65	[23]
	Cytoplasmic resistance, R_{cy} ($k\Omega$)	149	*
	Membrane capacitance, C_{mem} (pF)	1.43	*

Continued on next page

Table 2.1 – continued from previous page

Parameter		Value	Reference
B-lymphocytes	Diameter, D_c (μm)	6.58	[23]
	Specific membrane capacitance, C_{sm} ($\mu F/cm^2$)	1.26	[23]
	Cytoplasmic conductivity σ_{cy} (S/m)	0.73	[23]
	Cytoplasmic resistance, R_{cy} ($k\Omega$)	133	*
	Membrane capacitance, C_{mem} (pF)	1.71	*
Macrophages	Diameter, D_c (μm)	23	[24]
	Specific membrane capacitance, C_{sm} ($\mu F/cm^2$)	2.44	[24]
	Cytoplasmic conductivity σ_{cy} (S/m)	0.37	[24]
	Cytoplasmic resistance, R_{cy} ($k\Omega$)	75	*
	Membrane capacitance, C_{mem} (pF)	40.55	*
PBS 1x solution	Conductivity, σ_s (S/m)	1.7	[25]
	Solution resistance with no cell ($k\Omega$)	88	*
	Solution resistance with a neutrophil ($k\Omega$)	103	*
	Solution resistance with a T-lymphocyte ($k\Omega$)	92	*
	Solution resistance with a B-lymphocyte ($k\Omega$)	92	*
	Solution resistance with a macrophage ($k\Omega$)	-	*

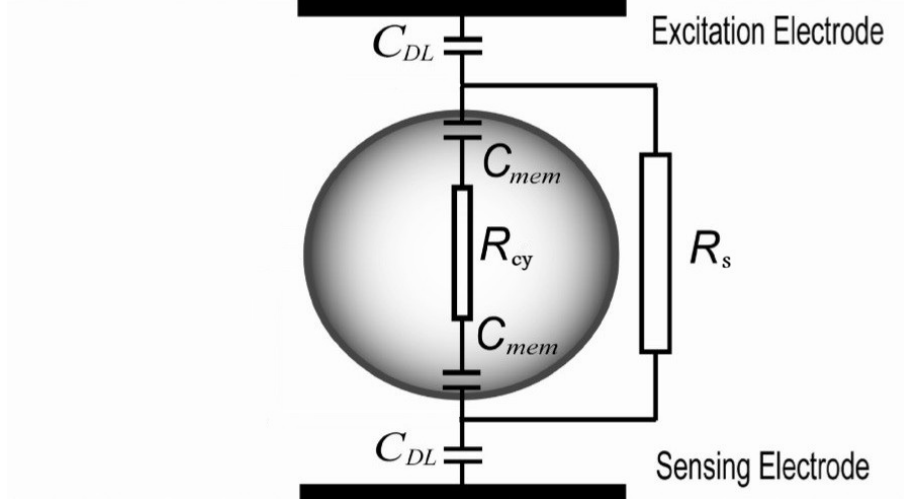


Figure 2.3: ECM used for impedance calculations [20]

TABLE 2.2: Calculated total impedance of cells in PBS buffer

Cell type	Frequency	Z_{out}
Neutrophils	300kHz	95.1k Ω
	1MHz	74.4k Ω
T-lymphocytes	300kHz	100.7k Ω
	1MHz	78.7k Ω
B-lymphocytes	300kHz	99.3k Ω
	1MHz	76.3k Ω
Macrophages	300kHz	118.7k Ω
	1MHz	87.8k Ω
No cell	300kHz	105.9k Ω
	1MHz	93.5k Ω

2.5 Measurement setup

In order for measurements to be clear and correct, a good measurement setup needs to be used. Multiple setups exist, specifically when it comes to the configuration

of electrodes. In this section, coplanar (Figure 2.3a) and facing (Figure 2.3b) electrode configurations are studied and compared to understand the advantages and disadvantages of each method.

According to Zhu et al. [14], the facing electrode configuration has a higher sensitivity and accuracy because of the more uniform electric field distribution. Coplanar electrode configuration suffers from more noise and a non-uniform electric field, which makes it more sensitive to cell position, but is simpler and more cost-effective to fabricate [14]. The chip fabricated by the University of Twente has coplanar electrodes (Figure 2.4). The electrodes are $10\mu m$ in length, spanning the whole channel in width and spaced $10\mu m$ apart. The microfluidic channel is $60\mu m$ in width and $65\mu m$ in height. However, as mentioned in the previous section, a facing electrodes configuration was assumed for the project to make the calculations easier. The parameters used can be found in Table 2.3.

There is double-layer capacitance present at the edge of the electrode and the liquid. It depends on the surface area of the electrode and capacitance per unit area C_d ($C_{DL} = A_{el} * C_d$). C_d is usually between $10\text{-}40 \mu F/cm^2$ [26]. For the project the assumed specific double-layer capacitance in the final system was $30 \mu F/cm^2$.

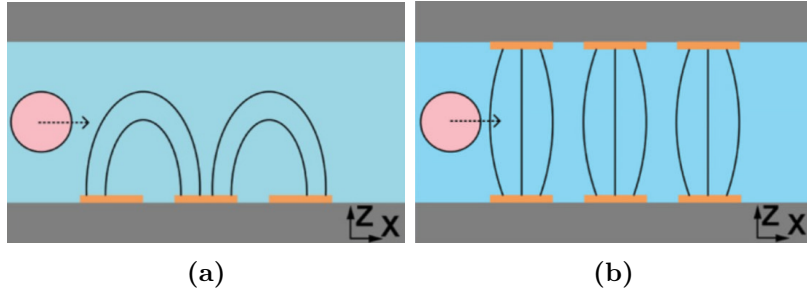


Figure 2.4: Coplanar (a) and facing (b) electrode configurations [14]

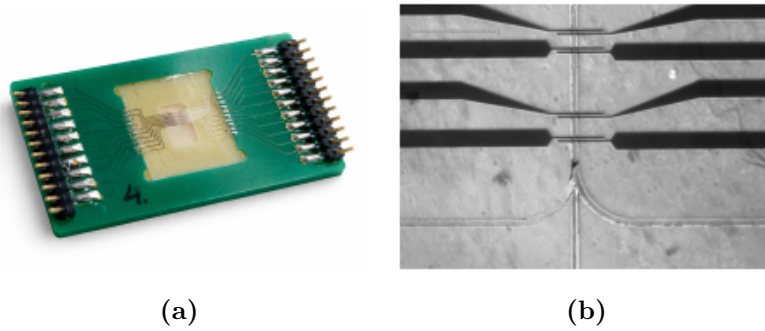


Figure 2.5: The chip fabricated by the University of Twente

TABLE 2.3: Parameters used for simulation circuit

	Parameter	Value	Reference
Electrode	Width (μm)	10	-
	Length (μm)	20	-
	Spacing, (μm)	30	-
	Capacitance per unit area, C_d ($\mu F/cm^2$)	30	[26]
	Double layer capacitance, C_{DL} (pF)	60	-

Different methods exist for improving sensitivity and accuracy of the measurements while reducing the noise. As an example, two experiments are considered, by Gawad et al. [27] with coplanar, and Gawad et al. [13] with facing electrodes. Both papers mention the use of differential measurements. By having three electrodes or two pairs of electrodes, in case of coplanar or facing configuration respectively, the difference of the reference (solution with no cell) and cell impedance can be calculated. This normalises the measurements against the reference level, which reduces noise, maximises sensitivity and can correct for uneven drift in electrode properties [14]. Besides differential measurements, lock-in amplifiers are used in both of the experiments. Lock-in amplifiers further help to extract the output signal from the noisy data. It uses phase-sensitive detection to isolate the component of the signal at a specific reference frequency and phase. The signal (Eq. 2.4) is multiplied with a reference (Eq. 2.5) and the output (Eq. 2.6) is passed through the low-pass filter to get a DC signal proportional to the amplitude of the original signal [28]. In case of $\omega_{sig} = \omega_{ref}$, the final signal is

$$V_{out} = \frac{1}{2} V_{sig} V_{ref} \cos(\theta_{sig} - \theta_{ref}) \quad (2.7)$$

$$V_{sig} \sin(\omega_{sig} t + \theta_{sig}) \quad (2.8) \quad V_{ref} \sin(\omega_{ref} t + \theta_{ref}) \quad (2.9)$$

$$\begin{aligned}
 V_{out} &= V_{sig} \sin(\omega_{sig} t + \theta_{sig}) * V_{ref} \sin(\omega_{ref} t + \theta_{ref}) \\
 &= \frac{1}{2} V_{sig} V_{ref} \cos([\omega_{sig} - \omega_{ref}]t + \theta_{sig} - \theta_{ref}) \\
 &\quad - \frac{1}{2} V_{sig} V_{ref} \cos([\omega_{sig} + \omega_{ref}]t + \theta_{sig} + \theta_{ref})
 \end{aligned} \quad (2.10)$$

The DC output of the lock-in amplifier also has an extra application. Furniturewalla et al. [29] write about using a lock-in amplifier to convert the high frequency output signal, to something that can be analysed by a cheap, low-resolution analog-to-digital converters (ADCs). This is particularly useful for developing a bedside diagnostic system as it opens up the possibility of using small and portable micro-controllers, such as Arduinos, for quick data processing.

2.6 Conclusion

In this chapter, an overview of the theoretical foundation necessary for the development of a bedside diagnostic system for SBP in cirrhosis patients was discussed. It covers SBP diagnosis, flow cytometry methods, dielectric properties of cells, ascitic fluids, equivalent circuit modelling and various measurement setups.

The main method for SBP diagnosis involves counting neutrophils in a sample of at least 1mL of ascitic fluids. If the number of neutrophils is $\geq 250/mm^3$, the diagnosis should be considered.

Current methods for counting neutrophils in ascitic fluids involve automated hematology analysers, but since they can be unreliable, manual count is often required, which leads to delays in the diagnosis. Among the available flow cytometry methods, electrical impedance spectroscopy provides an alternative label-free approach for single-cell analysis. It can be performed with small, portable devices, but has its limitations: low sensitivity and high signal noise.

The dielectric properties of cells, specifically the permittivity and conductivity and three dispersion regions, α , β and γ , provide the basis for cell differentiation using EIS and link the measurement frequencies to the physical properties of a cell, such as its size or membrane morphology. The β -dispersion range (0.1–3 MHz) was identified as the most appropriate for the SBP diagnosis system, as it allows cells to be distinguished based on multiple properties while avoiding the noise occurring at lower and higher frequencies. Equivalent circuit models help relate the total impedance measurements to the membrane and cytoplasm impedances, which can be used to distinguish different cell types.

The most prevalent cell types in ascitic fluids were neutrophils, lymphocytes and macrophages. The system needs to be able to distinguish between these cell types to accurately count the neutrophils.

Regarding the measurement setup, the facing electrode configuration offers higher sensitivity than the coplanar configuration, but is more difficult to fabricate. Differential measurement and lock-in amplification were identified as important setup elements for noise reduction and signal extraction. Additionally, the use of lock-in amplifiers also enables the use of low-cost microcontrollers, which greatly increases portability.

Overall, the findings provide a good basis for the design and development of an impedance-based flow cytometry system capable of counting neutrophils in ascitic fluid for rapid SBP diagnosis.

Chapter 3

Methodology

In this chapter the methodology used during the project is described.

3.1 Design process

For this project, Creative Technology design process by Mader and Eggink [30] was chosen. It consists of four phases: Ideation, Specification, Realisation and Evaluation (Figure 3.1). The process is very circular, during any of the phases it is common to go back to the previous ones if needed.

3.1.1 Ideation

During the ideation phase, background research is conducted to deepen the understanding of the problem. Additionally, an overview of similar technologies is created. Based on the findings, multiple ideas for the solution are generated.

3.1.2 Specification

In this phase, system requirements are defined based on the ideas and research done during the ideation phase.

3.1.3 Realisation

During the realisation phase, the system defined in the specification phase is constructed.

3.1.4 Evaluation

During the evaluation phase, the system is evaluated based on various factors. Those include functionality and usability. User testing is often an important part of this phase.

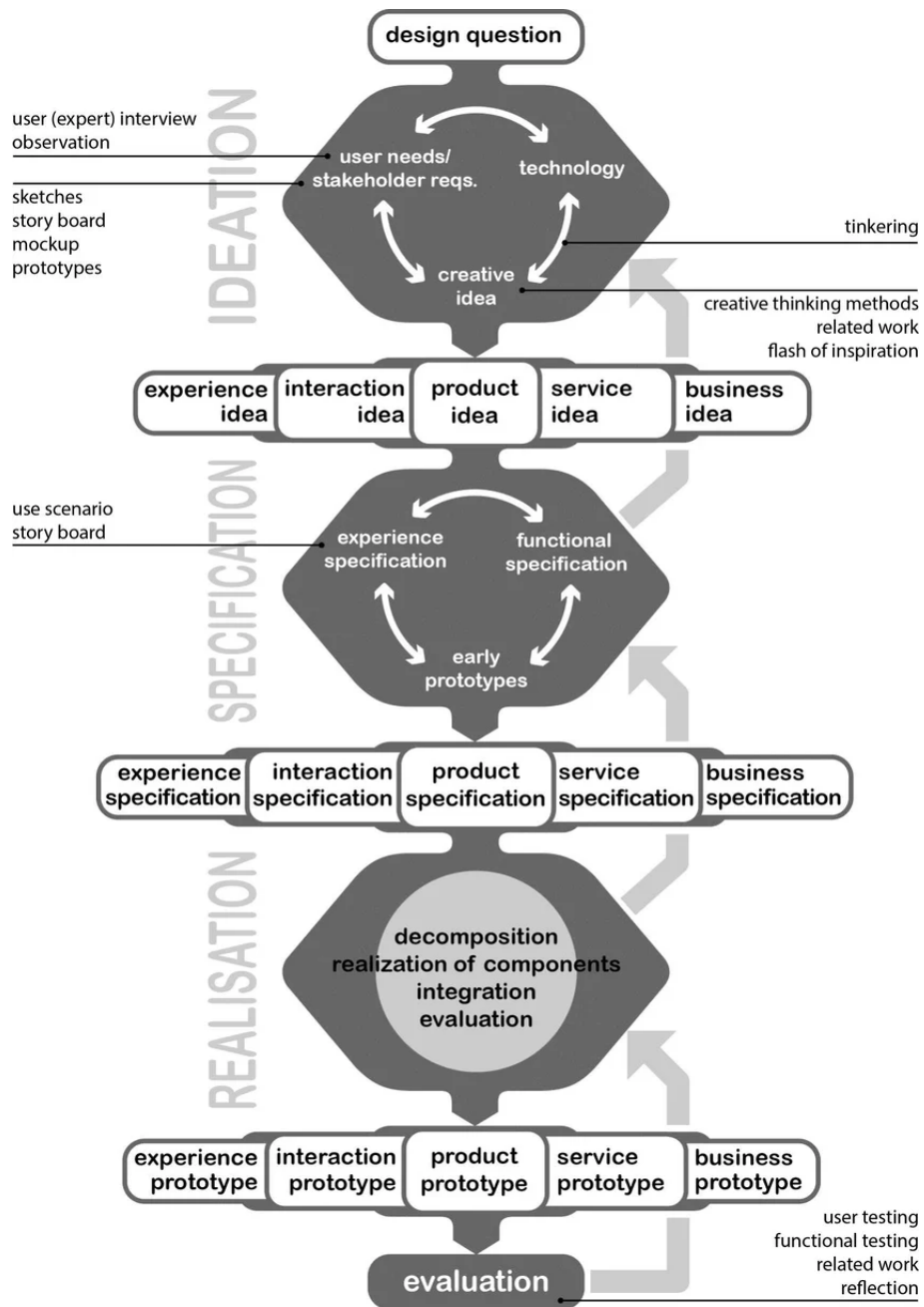


Figure 3.1: Creative Technology design process

3.2 Stakeholder analysis method

For the stakeholder analysis, a Power-Interest matrix was used [31]. The matrix consists of 4 quadrants:

- **High Power - High Interest:** Stakeholders that are directly involved in the project and who have a big impact on its success. Their expectations must be closely managed
- **High Power – Low Interest:** These stakeholders need to be kept satisfied despite them having little to no interest in the project because of their high power
- **Low Power – High interest:** Should be kept informed about the project's developments. They can often be helpful with the details of the project
- **Low power – Low interest:** Should be monitored, but without extensive communication

Chapter 4

Ideation

In this chapter, the stakeholders and their needs are described. Based on that, the requirements are formulated. Furthermore, different prototype concepts are shown and discussed.

4.1 Stakeholders

Multiple stakeholders were identified for the project. An overview of them is written in Table 4.1.

TABLE 4.1: Project stakeholders

Stakeholder	Description	Power	Interest
The author (Marko Hnatiuk)	Student conducting the research and developing the prototype	High - Main responsible for the development and in charge of most of the decisions	Interested in doing the research about the topic and developing a working prototype at the end of the project
The doctors	The hospital doctors who are the primary users of the final product	Medium - Their needs are the base for the requirements of the prototype. However, they do not directly take part in its development	Interested in having a more efficient method to diagnose SBP, leading to less workload

Continued on next page

Table 4.1 – continued from previous page

Stakeholder	Description	Power	Interest
Project supervisors (J. Berendsen & C. Salm)	Project supervisors guiding the student during the research and development phases	Medium - By providing guidance, they can influence some of the decisions	Interested in the successful completion of the project
Patients	Hospital patients with potential SBP	Low - They highly benefit from the product, but have no influence on the development process	The main beneficiary of the final product since it can lead to quicker diagnosis and treatment

4.2 Stakeholder analysis

Each of the stakeholders has a different level of influence on the project, depending on their power and interest in the project. Figure 4.1 shows where the aforementioned stakeholders lie on the power-interest matrix [31].

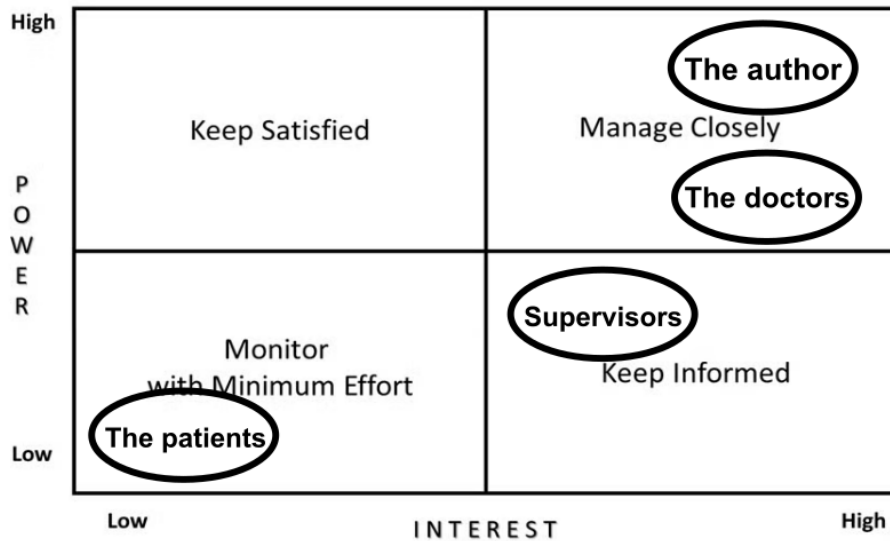


Figure 4.1: Stakeholder Power-Interest matrix [31]

4.3 Requirements

The requirements were formulated based on the needs of the doctors, stakeholders with the most power and interest, as well as the background research.

Dr. Kirsten Pondman, main contact person with the doctors at the University of Twente, was interviewed. The following system requirements were defined:

- The system must count all of the neutrophils in a sample within 10 minutes.

- The system must display the neutrophil count.
- The system should distinguish between neutrophils, lymphocytes and macrophages.
- The system must not emit bright light or make noise while not actively working.
- The system should take measurements at two frequencies, 300kHz and 1MHz

4.4 System concepts

Two system concepts are described in this section, as well as the final chosen concept.

4.4.1 Concept 1

The first concept of the impedance based cell-counting system is building one using dedicated impedance meters. Specifically, two devices were considered: Analog Discovery 2 and EmStat4S. Both devices are able to measure electrical impedance without much extra set up. Both are also capable of generating a signal, so they remove the need for an external function generator. However, EmStat4S can only measure at frequencies up to 200kHz, which is lower than required the 300kHz and 1MHz. Another problem with these devices is that they are not able to measure the electrical impedance at two different frequencies simultaneously. This means that at least two devices would be needed for the final system. With their price being 300€ - 400€ per device, the final system will be quite expensive. Since these devices have other functionalities that is not being utilised, using them would be overpaying for unnecessary features. They also require a laptop with a special software to operate, which further increases the costs and reduces portability. Due to these drawbacks, another concept was considered (see Concept 2).

4.4.2 Concept 2

The second concept is building a custom circuit using various components. The electrical impedance spectroscopy process consists of generating a signal that goes to the chip and the signal processing elements like a lock-in amplifier, regular signal amplifiers and filters. All of these can be built using simple components, op-amps and already existing integrated circuits. They only cost 5€ - 30€ depending on the component, which makes the final system very affordable. Additionally, the circuit can be built tailored to the specific requirements. Arduino can be used to analyse the signal and display the information on the screen. A lock-in amplifier turns the AC signal into DC signal, so the high frequency input wave would not be a problem for the Arduino. Since the circuit can be made into a printed circuit board, the final system can be made very compact and portable.

4.5 Chosen concept

In the end, the second concept, a custom circuit, was chosen, mainly due to the ability to customise it for the specific requirements and it being cheaper and more portable.

Chapter 5

Realisation

In this chapter, the final system is described. Multiple experiments were conducted, using an impedance spectroscope as well as the the built system. The results are compared and discussed.

5.1 Experiment setup

Impedance of neutrophils, T-lymphocytes, B-lymphocytes and macrophages are measured using two different methods: Analog Discovery 2 and the custom-built circuit with Arduino. The components are placed on a breadboard, which has an internal parasitic capacitance of about $6pF$, measured with Analog Discovery 2 (Figure 5.1). This causes the signal to become unreliable around $600-700kHz$, as can be seen in Figure 5.2. To avoid the effects of the parasitic capacitance, the further measurements were done at $300Hz$ and $1kHz$. The values of impedances in the ECM are scaled accordingly. Specifically, all the capacitances are multiplied by 1000. For both methods, the component values used for the ECM (Figure 5.3) are listed in the Table 5.1.

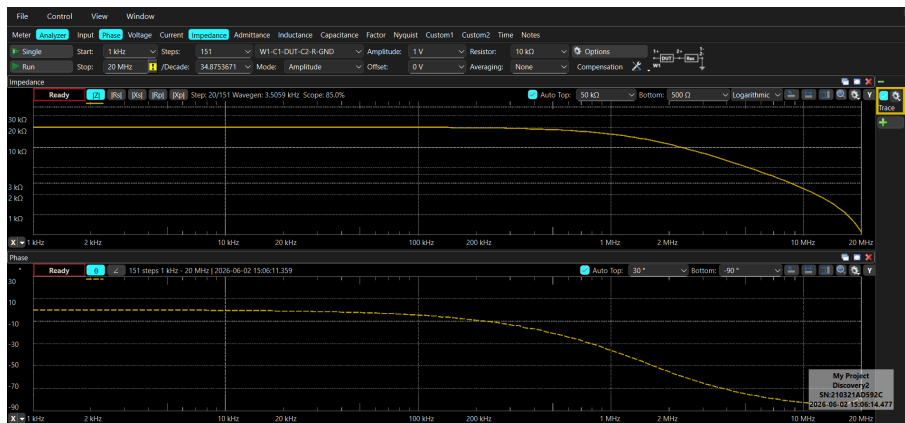


Figure 5.1: Frequency sweep of a breadboard. Impedance(top) and phase (bottom) graph. Equivalent to $6pF$ capacitance.

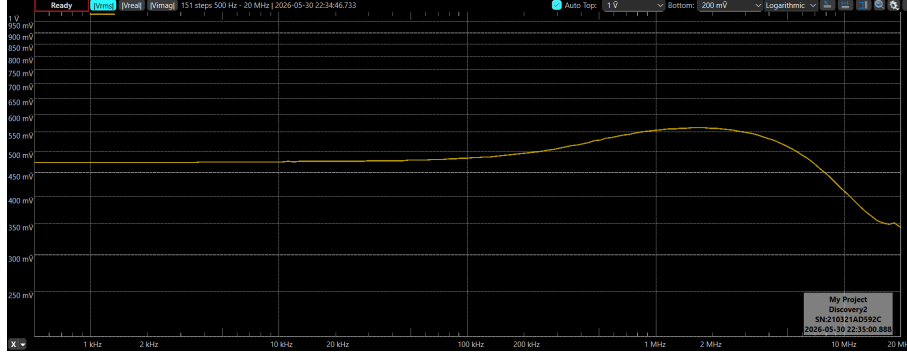


Figure 5.2: Voltage sweep of a breadboard

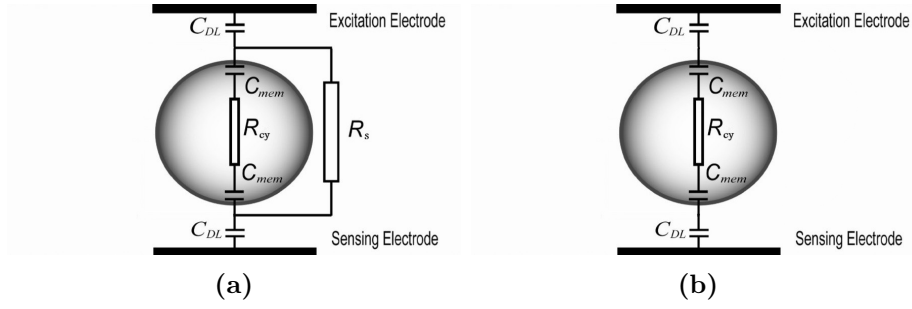


Figure 5.3: Equivalent circuit models used for the experiments. For macrophages (b) and for the other cells (a)

TABLE 5.1: Parameters used for simulation circuit

Parameter		Value
Neutrophils	Cytoplasmic resistance, R_{cy} ($k\Omega$)	168
	Membrane capacitance, C_{mem} (nF)	6.8
T-lymphocytes	Cytoplasmic resistance, R_{cy} ($k\Omega$)	147
	Membrane capacitance, C_{mem} (nF)	1.36
B-lymphocytes	Cytoplasmic resistance, R_{cy} ($k\Omega$)	131
	Membrane capacitance, C_{mem} (nF)	1.73
Macrophages	Cytoplasmic resistance, R_{cy} ($k\Omega$)	75
	Membrane capacitance, C_{mem} (nF)	42

Continued on next page

Table 5.1 – continued from previous page

Parameter		Value
PBS 1x solution	Solution resistance with no cell, R_s ($k\Omega$)	88
	Solution resistance with a neutrophil, R_s ($k\Omega$)	100
	Solution resistance with a T-lymphocyte, R_s ($k\Omega$)	93
	Solution resistance with a B-lymphocyte, R_s ($k\Omega$)	93
	Solution resistance with a macrophage, R_s ($k\Omega$)	-
Electrode	Double layer capacitance, C_{DL} (nF)	68

5.2 Analog Discovery results

Using Analog Discovery 2, an impedance frequency sweep was done from $f = 1Hz$ to $f = 5MHz$ for the 4 cell types and in a "no cell" situation. The results were meant to be used as a baseline for the evaluation of the custom built circuit.

5.2.1 Results

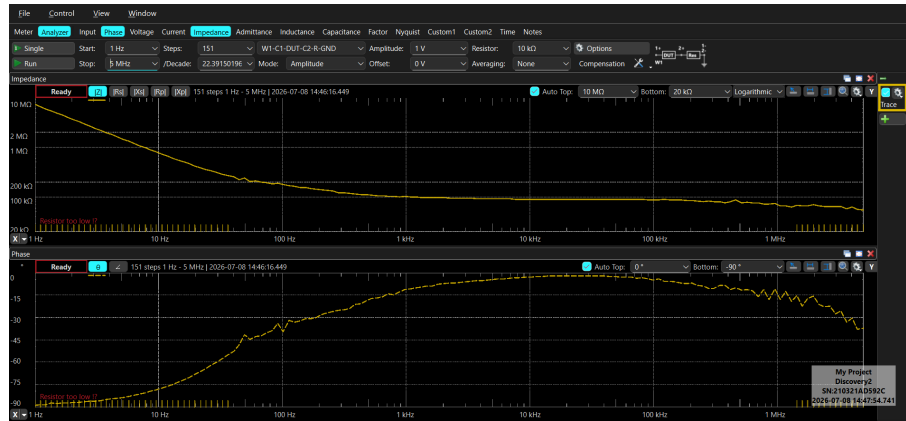


Figure 5.4: Neutrophil frequency sweep impedance(top) and phase (bottom) graph

Impedance values at frequencies of interest (316Hz and 982Hz): $Z_{316} = 118.3k\Omega$, $Z_{982} = 99.5k\Omega$

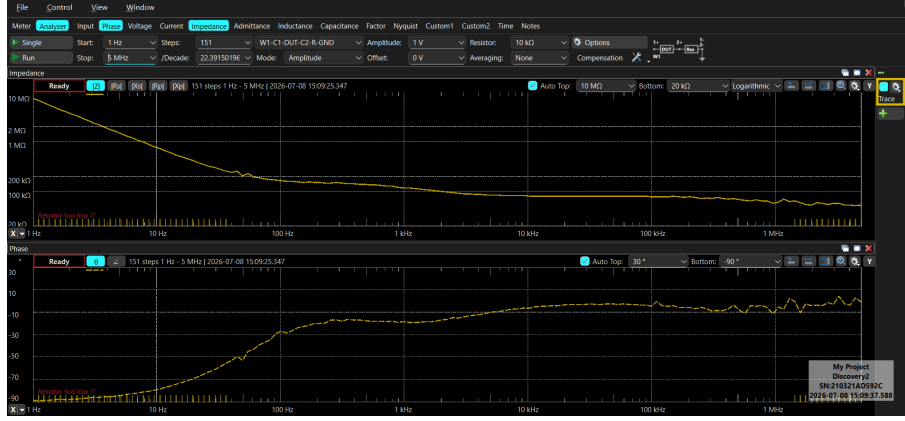


Figure 5.5: T-lymphocyte frequency sweep impedance(top) and phase (bottom) graph

Impedance values at frequencies of interest (316Hz and 982Hz): $Z_{316} = 142.3k\Omega$, $Z_{982} = 116.2k\Omega$

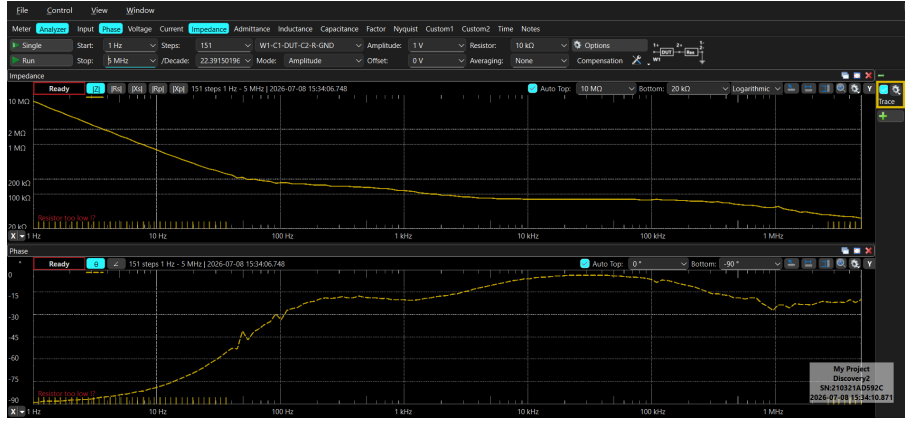


Figure 5.6: B-lymphocyte frequency sweep impedance(top) and phase (bottom) graph

Impedance values at frequencies of interest (316Hz and 982Hz): $Z_{316} = 140.4k\Omega$, $Z_{982} = 115.9k\Omega$

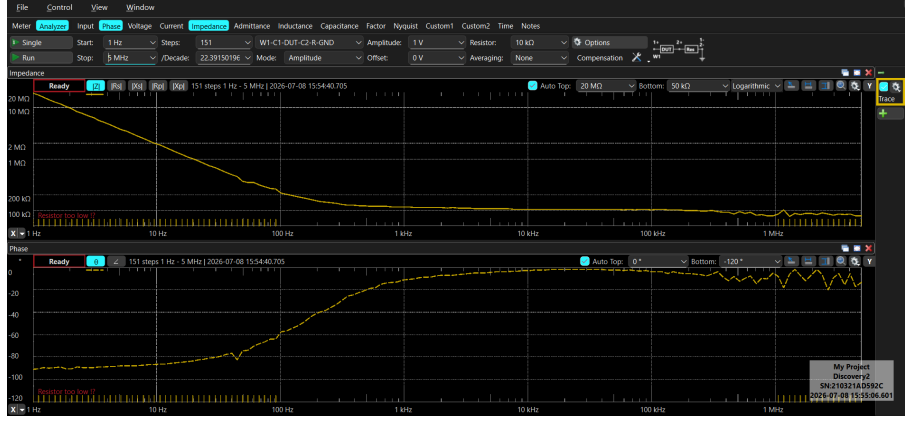


Figure 5.7: Macrophage frequency sweep impedance(top) and phase (bottom) graph

Impedance values at frequencies of interest (316Hz and 982Hz): $Z_{316} = 130.1k\Omega$, $Z_{982} = 115.2k\Omega$

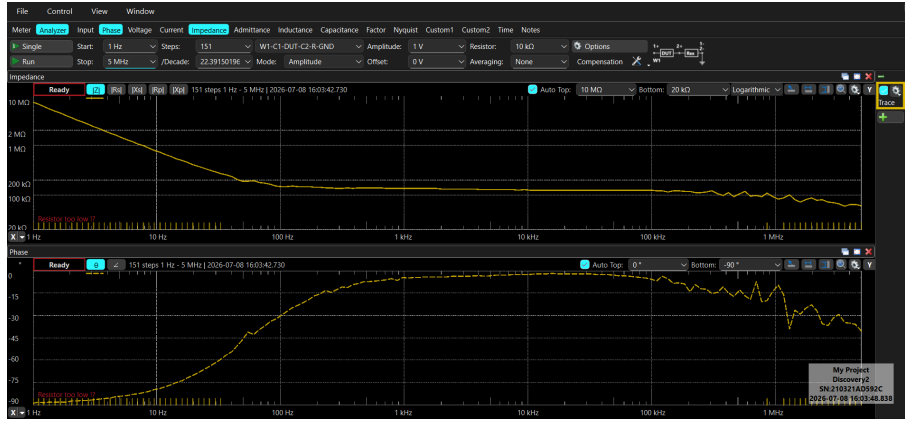


Figure 5.8: No cell frequency sweep impedance(top) and phase (bottom) graph

Impedance values at frequencies of interest (316Hz and 982Hz): $Z_{316} = 134.2k\Omega$, $Z_{982} = 133.9k\Omega$

5.2.2 Discussion

Out of the frequencies present in the frequency sweep data, the frequencies 316Hz and 982Hz were chosen for the observation since they are the closest to the 300Hz and 1000Hz.

For all of the experiments, the impedance up until around 25Hz is dominated by C_{DL} . At that point, Z_m is still relatively high, so the total impedance is mostly represented by R_s . The membrane polarises at around 1000Hz, and C_m starts to influence the total impedance. The drop in impedance at around 1MHz is due to the parasitic impedance in the breadboard, which can be modelled as a parallel capacitor to the ECM.

Looking at a measured impedance of cells, a clear difference can be seen at both 316Hz and 982Hz. Since the main purpose of the system is to count neutrophils,

only the difference between measured Z_{neut} and other cells will be examined. The difference among the cells, other than neutrophils is not as important, so it will not be discussed.

At 316Hz, Z_{neut} is 16.9% ($24k\Omega$) lower than $Z_{T-lymph}$, 15.7% ($22.1k\Omega$) lower than $Z_{B-lymph}$, 9.1% ($11.8k\Omega$) lower than $Z_{macroph}$ and 11.8% ($15.9k\Omega$) lower than $Z_{no-cell}$

At 982Hz, Z_{neut} is 14.4% ($16.7k\Omega$) lower than $Z_{T-lymph}$, 14.2% ($16.4k\Omega$) lower than $Z_{B-lymph}$, 13.6% ($15.7k\Omega$) lower than $Z_{macroph}$ and 25.7% ($33.4k\Omega$) lower than $Z_{no-cell}$

The difference is bigger than the calculated difference (table 5.2), except for with macrophages at 300Hz.

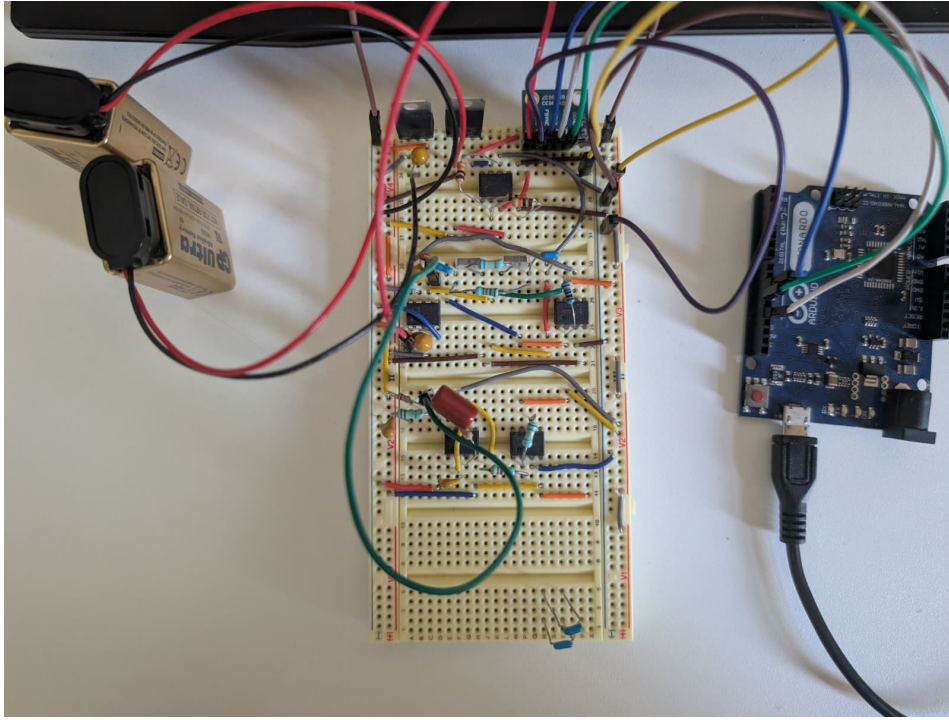
TABLE 5.2: Calculated total impedance of cells in PBS buffer

Cell type	Frequency	Z_{out}	$\frac{Z-Z_{neut}}{Z} * 100\%$
Neutrophils	300Hz	95.1k Ω	-
	1kHz	74.4k Ω	-
T-lymphocytes	300Hz	100.7k Ω	5.6%
	1kHz	78.7k Ω	5.5%
B-lymphocytes	300Hz	99.3k Ω	4.2%
	1kHz	76.3k Ω	2.5%
Macrophages	300Hz	118.7k Ω	19.9%
	1kHz	87.8k Ω	15.3%
No cell	300Hz	105.9k Ω	10.2%
	1kHz	93.5k Ω	20.4%

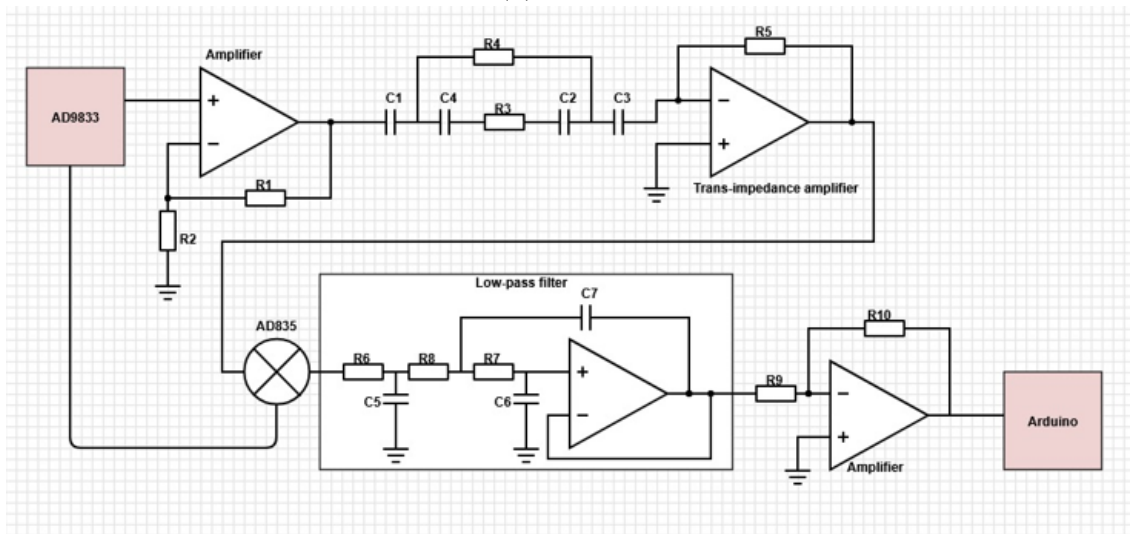
5.3 Arduino with a custom built circuit

The circuit was built on a breadboard. Figure 5.9 shows the circuit (a) and the circuit diagram (b). An excitation signal is generated using AD9833. It can generate signals up to 12.5MHz, which makes it suitable for the application of the project. However, the maximum amplitude of the AD9833 generated signal is 0.65V. The excitation signal amplitude is directly proportional to SNR [4], so the AD9833 output is fed into an 3x op-amp voltage amplifier. The signal then goes to an ECM of a

cell (electrodes of the microfluidic chip in the future). A trans-impedance amplifier converts the current from the ECM of a cell into a voltage, which goes to the AD9833 mixer. It multiplies the signal with the reference, which is the original AD9833 output. Eq. 5.1 shows the signal after the multiplication. This signal goes into a 3rd order active Sallen-Key Low-pass Filter with 10Hz cut-off frequency. The cut-off frequency was chosen as low as possible to attenuate the most noise. In situations where the cells are flowing through the channel, the cut-off frequency needs to be high enough to pass the peaks in the signal from the passing cells. This is discussed more in-depth in the next chapter.



(a)



(b)

Figure 5.9: Circuit built on a breadboard (a) and circuit diagram (b)

The filter removes the AC element of the AD835 output which leaves only the low DC component, proportional to the cell's signal amplitude (Eq. 5.2). Finally, after a 38x op-amp amplifier, the signal goes to the Arduino.

$$\begin{aligned}
V_{out} &= V_{sig} \sin(\omega t + \theta_{sig}) * V_{ref} \sin(\omega t + \theta_{ref}) \\
&= \frac{1}{2} V_{sig} V_{ref} \cos(\theta_{sig} - \theta_{ref}) \\
&\quad - \frac{1}{2} V_{sig} V_{ref} \cos(2\omega t + \theta_{sig} + \theta_{ref})
\end{aligned} \tag{5.1}$$

$$V_{out} = \frac{1}{2} V_{sig} V_{ref} \cos(\theta_{sig} - \theta_{ref}) \tag{5.2}$$

The circuit is being powered using 2 9V batteries, connected to L7805 and L7905. This provides a +5V and -5V power rails, that power op-amps, AD9833 and AD835.

Table 5.3 gives a list of the components used in the circuit. Resistors and capacitors are not included, since different values can be used to reach the same amplification ratio for the amplifiers and cut-off frequency for the filter.

TABLE 5.3: Components used for the system

Component	Used for	Description
NE5534AP	Excitation signal amplifier	Provides a 2x amplification
	Trans-impedance amplifier	Transforms current signal into a voltage with a ratio -20000:1
	DC signal amplifier	Amplifies the low-pass filter output by -38x to make it more detectable by the Arduino
AD9833	Signal generation	Generates the excitation signal
AD835	Signal multiplication	Part of the lock-in amplifier. Multiplies the signal with the original wave reference
L7805	Power supply	Converts 9V into 5V
L7905	Power supply	Converts 9V into -5V
Arduino Leonardo	Signal processing	Measures the final signal and is used for counting the cells

5.3.1 Results

TABLE 5.4: Arduino measurements

Cell type	Frequency	Analog signal	Z_{out}	$\frac{Z-Z_{neut}}{Z} * 100\%$
Neutrophils	200Hz	758	65.8k Ω	-
	300Hz	793	62.3k Ω	-
	500Hz	834	59.8k Ω	-
	1kHz	860	58.0k Ω	-
	3kHz	864	57.7k Ω	-
T-lymphocytes	200Hz	752	66.3k Ω	0.75%
	300Hz	763	65.3k Ω	4.59%
	500Hz	776	64.2k Ω	6.85%
	1kHz	804	62.0k Ω	6.49%
	3kHz	866	57.6k Ω	-0.17%
B-lymphocytes	200Hz	754	66.1k Ω	0.45%
	300Hz	765	65.2k Ω	4.45%
	500Hz	779	64.0k Ω	6.56%
	1kHz	813	61.3k Ω	5.38%
	3kHz	868	57.4k Ω	-0.52%
Macrophages	200Hz	716	70.0k Ω	6.00%
	300Hz	762	65.4k Ω	5.74%
	500Hz	799	62.4k Ω	4.17%
	1kHz	820	60.8k Ω	4.61%
	3kHz	827	60.3k Ω	4.31%
No cell	200Hz	774	64.4k Ω	-2.17%
	300Hz	785	63.5k Ω	1.89%
	500Hz	789	63.2k Ω	5.38%
	1kHz	789	63.2k Ω	8.23%
	3kHz	792	62.9k Ω	8.27%

Table 5.4 shows the measurements done with the custom-built circuit and the Arduino. Z_{out} - measured impedance of a cell (or solution when no cell is present) is calculated with Eq.5.3, where V_{cell} is the amplitude of signal over the cell, $R_{trans-imp}$ - the value of the resistor in trans-impedance amplifier, which sets the multiplication factor between the input current and the output voltage, V_{ref} - the amplitude of the AD835 reference, n - the multiplication factor of the final amplifier and A is the analog reading signal of the Arduino.

$$Z_{out} = \frac{V_{cell} * R_{trans-imp} * V_{ref} * n * 1024}{A * 5} \quad (5.3)$$

5.3.2 Discussion

Looking at the Arduino measurements, the first thing that is noticeable is that the measured impedance values are drastically different from those measured with Analog Discovery 2. The explanation for it can be that all of the wires and components in the circuit introduce extra parasitic capacitance and/or loading effects which cause the output to be different.

When it comes to the difference between neutrophils and other cells, it is around 4-6%, with some outliers. This is less than Analog Discovery 2 results, but it is in the range of other studies [27], [3]. Besides 300Hz and 1kHz, measurements were also taken at 3 additional frequencies: 200Hz, 500Hz and 3kHz. This was done to see if other frequencies will lead to better results. Table 5.4 lists the mean absolute difference in percentage among the cells for every frequency. It shows that the biggest differences are at 500Hz and 1kHz. This means that the system is better at distinguishing cells at these frequencies as opposed to the previously chosen 300Hz and 1kHz.

TABLE 5.5: Mean absolute difference by frequency

Frequency	Mean absolute difference
200Hz	2.34%
300Hz	3.92%
500Hz	5.74%
1kHz	6.17%
3kHz	3.32%

Chapter 6

Discussion and Recommendations

In this chapter, the performance and limitations of the system are discussed and recommendations for future work are given.

6.1 Performance and limitations

Measurement results using the custom built-circuit with Arduino show around 6% difference in impedance at 500Hz and 1kHz. This is in line with other studies [27], [3] which indicates that it can be used for detecting and distinguishing cells in ascitic fluids. However, these results have some limitations that are important to address while discussing the viability of the system.

First, the ECM of the cells used for impedance measurements is based on a parallel electrode configuration and the assumption that the electric field between the electrodes is uniform. This allows for easy calculations, but in reality, the chip has coplanar electrodes with a highly non-uniform electric field. In the future, ECM for coplanar electrodes can be made using the finite element model method with a tool like COMSOL or similar.

Secondly, the circuit was built on a breadboard, using wires and bulky components. Even though the length of the wires was kept to a minimum, such setup leads to parasitic inductance and capacitance. And when the change in a signal between cells is only 5%, even a small parasitic effects can lead to wrong results. The next step should be to build the system on a printed circuit board (PCB). This should limit the influence of parasitic effects which can improve sensitivity.

Finally, the system was tested in a static scenario, where a cell was modelled using resistors and capacitors. In reality, the cells will be flowing through the channel, and the signal will be constantly changing. It is hard to predict how the current system will perform in that case. It is likely that additional filters and amplifiers will need to be implemented. It is also important to keep in mind that even though the signal that goes into the Arduino is DC, when the cells will be flowing through the channel the signal will show a lot of peaks, with a frequency proportional to the flow speed. The maximum frequency of the peaks is limited by the sampling rate of a microcontroller's ADC. Taking a $F=5\mu m/min$ flow and a channel of $60\mu m$ width and $65\mu m$ height (area $A=3900\mu m^2$), the speed of a cell is $v = \frac{F}{A} = 21 mm/s$. With a gap between electrodes being $10\mu m$, and electrode width of $10\mu m$, the time

a cell spends in the sensing area is at most $t = \frac{d+2w}{v} = 1.4 \text{ ms}$. Arduino Leonardo has a maximum sample rate of 9615 samples/sec , so each peak will have at most $n = t * 9615 = 13$ samples. This might not be enough samples for an accurate cell detection. A solution can be using a different ADC, such as ESP32, or lowering the flow rate.

Additionally, the cut-off frequency of the lock-in amplifier's low pass filter needs to be high enough to not block the peaks of passing cells. With $5\mu\text{m}$ of flow speed, the peaks' frequency is $f = \frac{1}{t} = 714\text{Hz}$, so to be safe, cut-off frequency needs to be at least 1kHz .

Chapter 7

Conclusion

A proof of concept of a portable system for SBP diagnosis using neutrophil count in ascitic fluids with impedance spectroscopy showed potential. The system was able to accurately distinguish between neutrophils and the other cells in the ascitic fluids, such as lymphocytes and macrophages.

An analog circuit was built on a breadboard, using AD9833 as a signal generator, AD835 combined with a low-pass filter as a lock-in amplifier and Arduino Leonardo as a signal processing unit. The circuit was tested at multiple frequencies with ECM of 4 different cell types, including neutrophils. The frequencies that showed the biggest difference in measured impedance between the cells were 500Hz and 1kHz. At these frequencies, neutrophils impedance was 5-6% different from that of the other cell types. This is in line with the difference that other studies report.

However, the current system has some limitations, which require additional research in order to make it usable in hospitals. The ECM was modelled using a facing electrodes configuration, while the existing chip has coplanar configuration. Additionally, the system was not tested in a scenario where cells are moving through the channel and the sensing area.

The low cost and high portability of the system is its main advantage. Overall, the project lays a good foundation for a future research into the topic as well as further advancements of a bed-side system for SBP diagnosis using impedance spectroscopy.

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